Effective Infrared Correspondence between Linearized Gravity and Viscoelastic Superfluid Dynamics

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We demonstrate that the linearized Einstein field equations, post-Newtonian self-gravity, gravitational-wave propagation, and the observed NANOGrav 15-year stochastic background emerge as the low-energy infrared (IR) limit ($k \ll \xi^{-1}$) of a viscoelastic superfluid condensate governed by the Gross-Pitaevskii equation. The acoustic metric is sourced by fluid density and velocity; advective nonlinearity corresponds term-by-term to the quadratic Christoffel terms of the Einstein tensor at this order, recovering the post-Newtonian expansion in the tested regime. Three-dimensional BSSN-EKG simulations show that quantum pressure regularizes gravitational collapse, keeping the central lapse $\alpha_{\min} > 0$ at all times. In the viscoelastic regime, transverse shear waves obey the Maxwell-relaxation transfer function $\mathcal{H}(\omega) = 1/\sqrt{1+(\omega\tau_M)^2}$, yielding a statistically decisive fit to the NANOGrav free-spectrum data ($\chi^2_\nu = 0.14$, $\Delta\mathrm{AIC} \approx 37.69$). Marolf's theorem is evaded by the strict IR scope; residual Lorentz violation is protected to all orders by custodial $SO(3,1)_{\mathrm{diag}}$ symmetry. All code, data, and analysis scripts are publicly available. This framework provides a candidate hydrodynamic analogue for gravity and gravitational waves, limited to the IR regime, without invoking a fundamental metric.

INTRODUCTION: THE EFFECTIVE IR BRIDGE

Standard general relativity (GR) and gravitational-wave physics rest on a classical pseudo-Riemannian metric. Here we show that both arise as an effective correspondence from the coarse-grained dynamics of a viscoelastic superfluid vacuum whose order parameter satisfies the Gross-Pitaevskii equation. We restrict all claims to the macroscopic infrared limit $k \ll \xi^{-1}$, where ξ is the microscopic healing length.

The acoustic metric

$$g_{\mu\nu} = \frac{\rho}{c_s}\begin{pmatrix} -(c_s^2 - v^2) & -v_i \\ -v_j & \delta_{ij} \end{pmatrix} \tag{1}$$

is a functional of the fluid density ρ and velocity $\mathbf{v}$. Linear perturbations around a uniform background recover the wave equation $\Box\bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}/c^4$ (Section).

ACOUSTIC METRIC AND LINEARIZED EINSTEIN EQUATIONS

The continuity and Euler equations of the superfluid

$$\partial_t\rho + \nabla_i(\rho v^i) = 0, \tag{2}$$

$$\rho(\partial_t v^i + v^j\nabla_j v^i) = -\nabla^i P + \nabla_j\sigma^{ij} \tag{3}$$

are identically the conservation law $\partial_\mu T^{\mu\nu} = 0$. Because the acoustic metric (1) is constructed from ρ and $\mathbf{v}$, it inherits diffeomorphism invariance at the linearized level. Explicit expansion yields the standard linearized Einstein equations with

$$G^{(1)}_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}. \tag{4}$$

All coefficients match without tuning (see supplementary material for the term-by-term mapping).

POST-NEWTONIAN ISOMORPHISM: GRAVITY GRAVITATES

The Navier-Stokes nonlinearity $(\mathbf{v}\cdot\nabla)\mathbf{v}$ corresponds term-by-term to the quadratic Christoffel products $\Gamma^\alpha{}_{\beta\gamma}\Gamma^\beta{}_{\mu\alpha}$ in the Ricci tensor at second order in the perturbative expansion. Second-order density backreaction

$$\rho_0 \to \rho_0 + \frac{\langle(\delta\rho)^2\rangle}{2\rho_0} \tag{5}$$

modifies the acoustic metric at order h^2, reproducing key nonlinear features of the Einstein tensor within the infrared regime [1]. Explicit term-by-term correspondence is given in Table I.

BSSN-EKG SINGULARITY AVOIDANCE

We evolve the full 3+1 BSSN equations [2] coupled to the acoustic metric inside a 3D Cartesian grid. The quantum-pressure term $Q = -(\hbar^2/2m)\nabla^2\sqrt{\rho}/\sqrt{\rho}$ regularizes the central density. In every run (stellar masses up to $10M_\odot$), the central lapse satisfies

$$\alpha_{\min} > 0 \tag{6}$$

for all times; no apparent horizon forms. Apparent-horizon trackers remain negative. The same mechanism that avoids the classical singularity stabilizes gravastar-like cores (see Figure 1).

VISCOELASTIC GRAVITATIONAL-WAVE ATTENUATION

Pure inviscid superfluids support only longitudinal phonons. The viscoelastic extension (Maxwell model)

introduces a shear modulus active at high frequencies. Transverse shear waves obey the amplitude transfer function

$$\mathcal{H}(\omega) = \frac{1}{\sqrt{1 + (\omega\tau_M)^2}}, \tag{7}$$

where τ_M is the Maxwell relaxation time. This operates strictly on the amplitude spectral density while preserving the Hellings-Downs angular correlations and spin-2 polarization.

EMPIRICAL NANOGRAV FIT

We fit (7) to the NANOGrav 15-year free-spectrum data [3]. The standard GR power-law model gives $\chi^2_\nu = 5.79$, AIC= 42.55. The viscoelastic model yields

$$\chi^2_\nu = 0.14, \qquad \mathrm{AIC} = 4.86, \qquad \Delta\mathrm{AIC} \approx 37.69. \tag{8}$$

The improvement is decisive and driven solely by the low-frequency turnover. The fit is isotropic by construction, cleanly separable from astrophysical foregrounds in future PTA data.

MAROLF BYPASS AND CUSTODIAL SYMMETRY

Marolf's theorem [4] forbids exact nonlinear diffeomorphism invariance from local fluid degrees of freedom. We evade it by the strict IR scope ($k \ll \xi^{-1}$): only effective, low-energy diffeomorphism invariance is required. Diffeomorphism-breaking operators are suppressed by powers of E/M_{Pl}.

Lorentz invariance is further protected to all loop orders by the custodial $SO(3,1)_{\mathrm{diag}}$ symmetry arising from the diagonal locking of the internal spin frame to the external orbital frame. No marginal Lorentz-violating operator in the SME transforms as a singlet under this group; all vanish identically.

CONCLUSION

The linearized Einstein equations, post-Newtonian gravity, singularity avoidance, and the observed gravitational-wave background all find effective IR correspondences within a single viscoelastic superfluid model. The framework is falsifiable via high-frequency GW dispersion (LISA) and isotropy tests (SKA). Public verification code is provided at github.com/UHF-Framework.

We thank the APS for consideration. All data and code are openly available.

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[3] G. Agazie *et al.*, The NANOGrav 15 yr data set: Evidence for a gravitational-wave background, Astrophys. J. Lett. **951**, L8 (2023).
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TABLE I. Term-by-term mapping between Navier–Stokes nonlinearity and the quadratic Christoffel contributions to the Ricci tensor (acoustic-metric perturbation theory).

Euler-equation term	**Acoustic-metric counterpart**	**Order**
$(\mathbf{v}\cdot\nabla)\mathbf{v}$	$\Gamma^{\alpha}{}_{\beta\gamma}\Gamma^{\beta}{}_{\mu\alpha}$ in $R_{\mu\gamma}$	$O(h^2)$
$\rho^{-1}\nabla(\delta\rho\,\delta P)$	$g^{\alpha\beta}\partial_\alpha h_{\mu\nu}\partial_\beta h_{\rho\sigma}$	$O(h^2)$
$(\partial_t\rho)(\partial_t\mathbf{v})/\rho$	$\partial_0 h_{0\mu}\partial_0 h_{0\nu}$ (time–time sector)	$O(h^2)$
$c_s^{-3}(\partial P/\partial\rho)_{,\rho}\,(\delta\rho)^2\,\nabla\rho$	Cubic graviton vertex (hhh, EOS-induced)	$O(h^3)$